

STAT 637 - HW 1.6

RJ Cass

September 10, 2026

1

Propose a reasonable GLM. Specify $Y_i \overset{ind}{\sim} f(u_i), g(u_i) = \eta_i, \eta_i = \mathbf{x}^\top \beta$

(a)

$$Y_i \overset{iid}{\sim} \text{Bernoulli}(u_i)$$

$$\ln\left(\frac{u_i}{1-u_i}\right) = \eta_i$$

$$\eta_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{4i}$$

- Y_i is an indicator of whether or not the admitted student enrolled
- u_i is the probability that the admitted student enrolls
- β_0 is the intercept
- β_{1-4} are the corresponding model coefficients (distance from home, scholarship amount, undergrad GPA, attendance of an event)
- $x_{i,1-4}$ are the values for each individual student

(b)

$$Y_i \overset{iid}{\sim} \text{Poisson}(u_i)$$

$$\ln(u_i) = \eta_i$$

$$\eta_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i}$$

- Y_i is the count of birds in an area
- u_i is the expected average number of birds
- β_0 is the intercept
- β_{1-3} are the corresponding model coefficients (elevation, forest cover, temperature)
- $x_{i,1-3}$ are the values for each individual area

(c)

$$Y_i \stackrel{\text{iid}}{\sim} \text{Normal}(\mu_i, \sigma^2)$$

$$\mu_i = \eta_i$$

$$\eta_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i}$$

- Y_i is the weight of a tomato
- μ_i is the expected average weight of tomatoes
- σ^2 is the population variance
- β_0 is the intercept
- β_{1-3} are the corresponding model coefficients (fertilizer, water, greenhouse)
- $x_{i,1-3}$ are the values for each individual plant

(d)

In the tomato scenario, I think there could be an interaction between fertilizer treatment and amount of water used. The new systematic component would be: $\eta_i = \beta_0 + \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \beta_4 x_{1i} x_{2i}$, where the last term $\beta_4 x_{1i} x_{2i}$ includes the coefficient of the interaction, and the combination of fertilizer/water for that plant. This interaction answers the question of whether the effect of the amount of water given to a plant on the weight of tomatoes it produces changes depending on the fertilizer treatment used.

2

(a) **Essentially unchanged.** The idea of the interaction stays the same and it maintains the same representation in the systematic component.

(b) **Essentially unchanged.** Comparing models with and without interactions is common practice. While interpretations might shift a little bit, the idea is the same

(c) **Carries over but requires modification.** The concept of evaluating the β 's as expected changes per one unit increase in the value holds, what it's changing does not. In linear regression the link function is just the identity so the 'one unit change' directly applies to the output. In logistic, the link function is different so the β actually represents the impact a one unit change has on the probability π_i

(d) **Essentially unchanged.** I think this is just exactly what a predicted value is, the expected value of the output given the inputs, right?

(e) **Does not carry over.** While regular linear models should have normal residuals, due to the discrete nature of logistic regression (output is 1 or 0), the residuals are discretized and non-normal

(f) **Does not carry over.** Due to the random component being a Bernoulli, each Y_i has a different variance depending on the π_i that feeds into it.

(g) **Essentially unchanged.** Response can have values of 0 or 1, you better not be getting values that aren't 0 or 1. Predictions should always be within the support of the response variable. If not, you did something wrong.

(h) **Essentially unchanged.** This holds in any case I can think of. As with all models, it's only as good as the experimental design, and if there is something happening behind the scenes that wasn't captured, your model also won't be able to identify it.

3

(a) Number of emergency-room visits during the next year

1. $\mathbb{Z}^+$ (positive integers)
2. It's discrete
3. No, a normal distribution is not reasonable for counts
4. Poisson
5. Is this the number of visits one person makes to the ER, or is it the number of visitors to one ER in the year?

(b) Whether a customer defaults on a loan

1. 0, 1
2. Probably equal tailed distribution
3. No, this is discrete 0/1 values, not good for normal
4. Logistic
5. Do we want a predicted output of yes/no whether or not they do default, or an output of the probability that they default?

(c) Proportion of 100 planted seeds that germinate

1. Integers in $[0, 100]$
2. Equal tailed distribution
3. Nope, again, discrete outcomes
4. Binomial
5. What's the initial guess of how many will germinate? Are we expecting extreme values (like only 1 or 2, or all of them), or are we more towards the middle (30-70ish)

(d) Annual household income

1. $\mathbb{R}^+$ (positive reals, although I guess maybe someone could just be in so much debt we'd categorize it as negative? haha)
2. Probably very right skewed (hello wealth inequity)
3. No, due to the expected heavy tail, not a good fit for normal
4. LogNormal (nice heavy tails)
5. Maybe some EDA, check if maybe something like a Chi-Squared would better represent the tails

(e) Number of goals scored by a soccer team in a match

1. $\mathbb{Z}^+$ (positive integers)
2. It's discrete, but probably low numbers
3. No, a normal distribution is not reasonable for counts

4. Poisson
5. Are PKs counted the same, especially if they go to a shootout?

(f) A student's score on a 20-question multiple choice test

1. Integers in $[0, 20]$
2. Probably left tailed (most do ok, maybe one or two do poorly?)
3. Probably not if assuming discrete values
4. Binomial
5. Is partial credit available?

(g) Number of days from infection until recovery

1. $\mathbb{Z}^+$ (positive integers)
2. Long tail (some people might just take a really long time to recover)
3. No, discrete, and tails, does not lead to normal
4. Geometric
5. What determines recovery? And is this like, a months long process, or just a few days?

(h) Number of emails received by an employee during a workday

1. $\mathbb{Z}^+$ (positive integers)
2. Again, probably long right tails (the CEO probably gets a LOT more emails than an L1 or something)
3. Nope, distinct counts
4. Poisson
5. Will this include all emails (including responses), or just new email threads?

4

(a)

These are fundamentally different because the first model allows for all real values, whereas the second model, due to the Poisson random component, only allows discrete values. They have different support.

(b)

If $Y_i = 0$, the first model is undefined ($\ln(0)$ is undefined)

(c)

In the first model, the log-response ($\ln(Y_i)$) is assumed to be linear. In the second model, the log-Poisson parameter ($\ln(u_i)$) is assumed to be linear, not the response.

(d)

Poisson variance is equal to the mean, so as the conditional mean changes, so does the variance

(e)

If the expected counts are really high, the Poisson model would have a really high variance. Thus, if the expected distribution is a high count, but relatively low variance, the transformed normal might actually be a good fit (also far away from 0 to avoid any issues with boundaries)

5

(a) $Y_i \stackrel{ind}{\sim} N(\mu_i, \sigma^2), \mu_i = \beta_0 + \beta_1 x_i + \beta_2 z_i$

Plausible Study: Length of newborn snakes dependent on type of food provided to the mother, and quantity of water provided.

- Y_i : Length of a newborn snake
- x_i : Quantity of water provided
- z_i : Type of food provided
- Statement addressed by β_1 : For every one unit increase in the amount of water provided, how much does the length of a newborn snake change?

(b) $Y_i \stackrel{ind}{\sim} \text{Bernoulli}(\pi_i), \text{logit}(\pi_i) = \beta_0 + \beta_1 x_i + \beta_2 z_i$

Plausible Study: Whether or not a snake egg hatches correctly dependent on type of food provided to the mother, and quantity of water provided.

- Y_i : Whether or not the egg hatches correctly (yes/no)
- x_i : Quantity of water provided
- z_i : Type of food provided
- Statement addressed by β_1 : For every one unit increase in the amount of water provided, how much does the likelihood of the egg hatching correctly change?

(c) $Y_i \stackrel{ind}{\sim} \text{Poisson}(\mu_i), \log(\mu_i) = \beta_0 + \beta_1 x_i + \beta_2 z_i$

Plausible Study: Number of eggs laid by a snake dependent on type of food provided to the mother, and quantity of water provided.

- Y_i : Number of eggs laid
- x_i : Quantity of water provided
- z_i : Type of food provided
- Statement addressed by β_1 : For every one unit increase in the amount of water provided, how much does the expected number of eggs laid change?

(d)

They all have the same systematic component. The randomness in each model also comes from the random component.

(e)

They all have different link functions and random components. This means the interpretation of the 'linear' portion changes as well.

6

(a) Binary

- Response Variable: Whether a boxer wins their fight or not
- Observable Unit: Victory (yes/no)
- Plausible Explanatory Variables: Weight, Height, hours trained, jab strength
- Candidate Response Distribution: Bernoulli
- Candidate Link Function: Logistic
- GLM: $Y_i \stackrel{\text{iid}}{\sim} \text{Bernoulli}(\pi_i), \text{logit}(\pi) = \eta_i, \eta_i = \beta_0 + \beta_1 x_i + \dots$
- Question the model would answer: how can we best predict whether a boxer will win their match
- Reason the GLM might be inadequate: There could be interactions between variables which would require us to revisit the systematic component.

(b) Count

- Response Variable: Number of fish caught in 1 hour
- Observable Unit: A fish
- Plausible Explanatory Variables: Lure/bait type, location, time of day
- Candidate Response Distribution: Poisson
- Candidate Link Function: Log
- GLM: $Y_i \stackrel{\text{iid}}{\sim} \text{Poisson}(\pi_i), \ln(\pi) = \eta_i, \eta_i = \beta_0 + \beta_1 x_i + \dots$
- Question the model would answer: What factors most influence how many fish are caught in an hour
- Reason the GLM might be inadequate: If I'm really bad at fishing and don't catch anything haha. But actually, I imagine there would be heavy interactions between bait type and location, etc.

(c) Normal might be questionable

- Response Variable: Household Incomes in Utah
- Observable Unit: Income for one household
- Plausible Explanatory Variables: Education, Race, Industry
- Candidate Response Distribution: Normal
- Candidate Link Function: Identity
- GLM: $Y_i \stackrel{\text{iid}}{\sim} \text{Normal}(\mu_i, \sigma^2), \mu_i = \eta_i, \eta_i = \beta_0 + \beta_1 x_i + \dots$

- Question the model would answer: What demographic features are most correlated with higher incomes in Utah
- Reason the GLM might be inadequate: We would expect income to potentially have a large right tail. In that case, normal might not be the best fit (though we could truncate the data to only the lower 90% or something).

7

Logistic regression makes more sense in this situation because while the normal linear regression could provide likelihoods of a customer renewing, the logistic regression will actually directly provide a yes/no prediction as to whether the customer will renew. It is also likely that a normal linear regression model fit to these data will not be able to meet the required assumptions, where as the logistic regression should be able to meet required assumptions.